DIPARTIMENTO DI INGEGNERIA INFORMATICA, MODELLISTICA, ELETTRONICA E SISTEMISTICA

IoT Security Report: Aigital Wi-Fi Extender Analysis

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1 Introduction

In this report we will discuss our security analysis of a Wi-Fi extender produced by Aigital.

1.1 Device Informations

The Aigital Wi-Fi Extender is a device that allows to extend the coverage of a Wi-Fi network.
It was bought in 2019 and is still available for purchase on Amazon[1] and it is sold at a price of around 20 euros.
The device has a web interface for configuration and management, which is accessible through the local network on 192.168.10.253.

Figure 1: Web interface

1.2 Objectives

As we will see in the next sections, our main goal was to exploit the CVE-2023-30403[1], CVE-2023-30404[2] and CVE-2023-30405[3] vulnerabilities, which allow for login by-pass, remote code execution

[1] `https://amazon.it/dp/B0DR7ZT2CS`

and cross-site scripting respectively.

We also wanted to analyze the firmware of the device, in order to find other potential vulnerabilities and to understand how the device works.

Finally, we wanted to patch the CVEs compiling a custom firmware.

1.3 Tools Used

We used a variety of tools for our analysis, including:

- A digital multimeter for individuating the pins of the UART interface
- A USB-to-TTL adapter for analyzing the UART communication
- A CH341A programmer with a soic clip for dumping the EEPROM with flashrom
- Binwalk for firmware analysis
- Ghidra and Cutter for reverse engineering the firmware
- A custom script for patching the firmware

All of our codes are available on our GitHub repository[2].

[2] `https://github.com/Gory-git/progetto_IoT`

2 Analysis

2.1 Hardware

Opening the device we exposed the PCB, which is shown in the figure below.

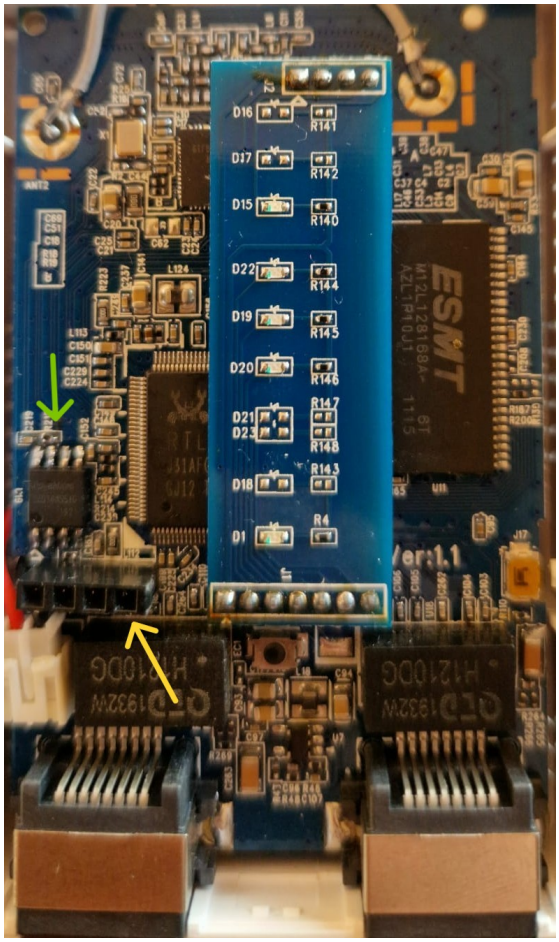

Figure 2: PCB of the Aigital Wi-Fi Extender (after de-soldering the eeprom and inserting a connector to make the UART easier to read)

By a visual inspection of the PCB we identified the pins of the UART interface, yellow arrow, the eeprom, green arrowand the following chips:

- chip 1
- chip 2
- chip 3

2.1.1 UART

Using a digital multimeter we individuated the pins of the UART interface, which are shown in the figure below, and are (from left to right):

- Pin 1: VCC (red arrow)
- Pin 2: TX (yellow arrow)
- Pin 3: RX (green arrow)
- Pin 4: GND (blue arrow)

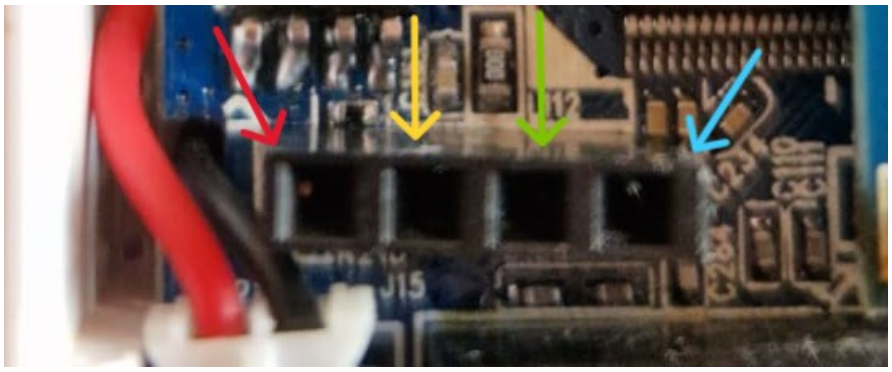

Figure 3: UART interface

Once we identified the pins of the UART interface, we found, with some patience, the correct baud rate for the communication, which is 38400 baud.
Booting the device and connecting to the UART interface with a USB-to-TTL adapter we were able to read the boot log of the device, which is shown in the figure below.

```
eth0 added. vid=9 Member port 0x1...
eth1 added. vid=8 Member port 0x10...
eth2 added. vid=9 Member port 0x2...
eth3 added. vid=9 Member port 0x4...
eth4 added. vid=9 Member port 0x8...
[peth0] added, mapping to [eth1]...
Realtek FastPath:v1.03
init started: BusyBox v1.13.4 (2013-12-29 13:17:23 CST)
Generated PIN = 00375528
Init Start...
Init bridge interface...

DO 8192E IQK !!!!
Done 8192E IQK !!!!

Init Wlan application...

set_wlanapp.c__setWlan_Applications__752_       wlMode:0

WiFi Simple Config v2.12.1-wps2.0 (2013.05.16-02:22+0000).

Register to wlan0
route: SIOCDELRT: No such process
iwcontrol RegisterPID to (wlan0)
IEEE 802.11f (IAPP) using interface br0 (v1.8)
Init WAN Interface...
□□□□□□□□!□□□□□□check ws ok

rlx-linux login: admin
Password:
Login incorrect
rlx-linux login:
```

Figure 4: Boot log of the device

Unfortunately, no U-Boot console is available, but the boot log is still very useful for us, since it allows us to identify rlx-linux as the operating system running on the device. The os we found, rlx-linux, is a custom open source Linux distribution for embedded devices.

We tried to submit in the login some common default credentials, unfortunately, we were not able to log in to the device.

2.1.2 Eeprom

Figure 5: EEPROM

2.2 Information Gathering

[4]

2.3 Software

2.3.1 Eeprom Dumping

2.3.2 Binary Analysis

2.3.3 Firmware Emulation

2.3.4 Firmware Analysis

3 Exploiting

3.1 Login By-pass

3.1.1 Exploitation

3.2 RCE

3.2.1 Executing Commands

3.2.2 Gaining Shell

3.2.3 Arp Poisoning

3.3 XSS

3.3.1 Exploitation

4 Fixing

4.1 Binary Patching

4.1.1 Patching Login By-pass

4.1.2 Patching RCE

4.1.3 Patching XSS

4.2 Compiling Firmware

5 Conclusion

References

[1] *CVE-2023-30403: Aigital Wi-Fi Access Point Login By-pass Vulnerability*. https://nvd.nist.gov/vuln/detail/CVE-2023-30403.

[2] *CVE-2023-30404: Aigital Wi-Fi Access Point Remote Code Execution Vulnerability*. https://nvd.nist.gov/vuln/detail/CVE-2023-30404.

[3] *CVE-2023-30405: Aigital Wi-Fi Access Point Cross-Site Scripting Vulnerability*. https://nvd.nist.gov/vuln/detail/CVE-2023-30405.

[4] *CVE-2023-30403, CVE-2023-30404 and CVE-2023-30405: Login By-pass, Remote Code Execution and Cross-Site Scripting Vulnerabilities found in Aigital Wi-Fi Access Point by Matteo Mandolini*. https://mandomat.github.io/2023-04-13-testing-a-cheap-wifi-repeater/.